

DSE Capacity vs Model Generation Workload

Purpose

The Data Science Environment, DSE, is being asked to support a different workload from the one it was originally sized for.

Successive model generations change the amount and type of processing required. The environment does not automatically scale when that workload changes.

DSE operated with **50 workers until February 2026**, when capacity increased to **100 workers**. That was a material uplift, but model development continues to evolve. The key question is whether DSE capacity is keeping pace with the workload now being introduced.

This page creates a single view of that relationship across model generations.

The challenge

Each model generation is different.

The number of features, the type of features, the amount of historical data, the volume of data processed, and the way the work has to be executed can all change.

That means the workload for OB4 is not automatically comparable with FE8, FV9 or FV10.

If the workload grows or becomes more complex while DSE capacity stays broadly fixed, the same constraints will keep reappearing.

The issue is therefore not one failed run or one difficult Feature Plan.

The issue is the gap between an evolving model-development workload and an environment that is upgraded less frequently.

What we mean by Feature Plan and Weekly Model Plan

Term	Simple explanation
Feature Plan	Generates feature values needed for a future model generation by running feature logic over historical data.
Weekly Model Plan	Refreshes the values of features already required by the established model estate using new weekly data and the required history.

The Feature Plan creates development demand. The Weekly Model Plan creates recurring demand.

Both place workload on DSE.

Why model generations create different workloads

A new model generation does not only mean a new model version.

The workload can change across several dimensions:

Workload dimension	What can change between generations	DSE impact
Feature volume	More candidate features need to be generated or evaluated	More calculations and larger outputs
Feature complexity	Features may use heavier aggregations, joins, mappings or behavioural logic	Higher compute and memory demand
Historical depth	Longer lookback periods may be required	More data must be read and processed
Data volume	The underlying customer or transaction population may be larger	Larger processing footprint
Execution pattern	Work may require splitting, reruns or multiple stages	More elapsed time and operational effort
Recurring workload	New feature generations can change what the Weekly Model Plan must maintain	Higher or different weekly demand

This is why final feature count alone is not enough to explain DSE workload.

For example, FV9 feature generation used approximately **three years of historical data** and full-volume processing encountered crashes, so the workload had to be split into smaller parts and later combined.

1. Capacity vs Model Generation

This is the headline view. It should show how the model workload changed alongside the DSE capacity available at the time.

Model generation / period	DSE workers	Workload evidence	Key constraint / observation
OB4	TBD	Feature Plan scale, history and processing pattern to confirm	TBD
FE8	TBD	Feature Plan scale, history and processing pattern to confirm	TBD
FV9	50 / transition to confirm	~3 years of history, partitioned processing	Full-volume runs encountered crashes
Feb 2026 capacity uplift	100	Capacity increased	Worker capacity doubled
FV10	100	Current Feature Plan workload to establish	Current constraint position to establish

What this should show

The comparison is not simply whether capacity increased.

It should show whether the increase in capacity has kept pace with the change in workload.

2. Feature Plan Workload Evolution

This view compares the development workload required to generate features for each model generation.

Measure	OB4	FE8	FV9	FV10
Features submitted for generation	TBD	TBD	TBD	TBD
Historical period processed	TBD	TBD	~3 years	TBD
Approx. data volume	TBD	TBD	TBD	TBD
Existing feature sets joined	TBD	TBD	FV8	TBD
Number of execution parts	TBD	TBD	Multiple	TBD
Reruns / failed executions	TBD	TBD	Observed	TBD
End-to-end generation time	TBD	TBD	TBD	TBD

The question this view answers

What did DSE have to process to generate the feature data required for each model generation?

This is more useful than comparing final feature counts alone.

3. Weekly Model Plan Evolution

This view compares the recurring workload DSE has to support once feature generations become part of the established model estate.

Measure	Model Plan V4	V5	V6	V7	V8
Feature estate supported	TBD	TBD	TBD	TBD	TBD
Number of features calculated	TBD	TBD	TBD	TBD	TBD
Historical data required	TBD	TBD	TBD	TBD	TBD
Typical weekly execution time	TBD	TBD	TBD	TBD	TBD
Peak execution time	TBD	TBD	TBD	TBD	TBD
Processing issues / constraints	TBD	TBD	TBD	TBD	TBD

The question this view answers

Is the recurring workload of the current Weekly Model Plan still comparable with the workload DSE was supporting when earlier capacity assumptions were made?

4. Final Model Evolution

This view provides the outcome of each model generation.

Measure	OB4	FE8	FV9	FV10
Final feature count	TBD	TBD	TBD	TBD
Stable VDR	TBD	TBD	TBD	TBD
Stable FPR	TBD	TBD	TBD	TBD
Features considered during development	TBD	TBD	TBD	TBD

This is useful context, but it is not a direct measure of DSE workload.

A model may finish with a similar number of features while requiring a much larger or more complex development workload to get there.

Why this will keep happening

Model development evolves continuously. Capacity changes happen at specific points in time.

That creates a repeating pattern:

Capacity is set

-> **Model workload changes**

-> **DSE absorbs the new demand**

-> **Constraints appear**

-> **Capacity or processing is adjusted**

-> **The next model generation changes the workload again**

Without a clearer link between model-generation planning and DSE capacity planning, the same issue is likely to return.

What this should support

The completed view should help us make an informed decision across three areas:

Area	Decision
DSE capacity	Does the environment have enough compute and headroom for the current and next model generation?
Workload optimisation	Can Feature Plans and Weekly Model Plans be made more efficient before adding more capacity?
Model-generation cadence	Should the sequencing or pace of model generations change where DSE capacity is constrained?

These are linked decisions. The answer may involve more than one.

Key message

DSE capacity has increased, but model-generation workload continues to change. The issue is whether the environment is evolving quickly enough to support the scale and complexity of the work now being introduced. If capacity, workload design and model-generation planning stay disconnected, the same constraints will keep returning.